Checkpoint 2: One-Page Project Plan

Big Data Project Work | Team 5 | Due Tuesday 3 November 2026

Team: Boomikaa P, Chahatleen Kaur, Saba Gul, Vanshika Malviya, K. Shreya, Tushar Rathor. **Job:** Data Engineer (Data Modeling), IBM Consulting.

1. Business problem

A UK online gift-ware shop (many customers are wholesalers) cannot trust its sales numbers. In the raw Online Retail II file about 23% of lines have no customer number, cancelled invoices (starting with C) are mixed with real sales, and the same product code can carry different descriptions. A report built straight from this file can show the wrong top customer, best-selling product or revenue per country, so marketing and stock money goes to the wrong places. We will fix the data, keep one clean record per product and customer, organise it in a star layout, and only then report on it. This is the work IBM asks of a Data Engineer (Data Modeling).

Decision the dashboard will support: which customers, products and countries the shop should focus on next year.

2. Public dataset and licence

Item	Detail
Dataset	Online Retail II, UCI Machine Learning Repository
Content	1,067,371 sales lines, 1 Dec 2009 to 9 Dec 2011: invoice, product code, description, quantity, date, price in pounds, customer number, country
Licence	CC BY 4.0: we may use and change the data if we give credit
Credit	Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D
Getting it in	The file is an Excel workbook, so we convert it to CSV on a laptop and upload it to a Unity Catalog Volume, because notebooks have limited internet access

3. Pipeline steps, each tied to a gap

#	Step (on Databricks)	Gap it closes	Who
1	Bronze: load the raw file exactly as it is, with a record of row counts	Moving and cleaning data (ETL/ELT)	Chahatleen, K. Shreya, Tushar
2	Silver: clean it: split out cancellations, fix text and country names, flag rows with no customer number	Keeping data correct and consistent	Chahatleen, K. Shreya, Saba
3	Draw the logical model in a free tool (for example draw.io)	Gap 1: design tool like ERWin; Gap 3: designing how data is organised	Boomikaa, Vanshika, Tushar
4	Write the table definitions (physical model) and build the gold star layout: one sales table in the middle with customer, product, date and country tables around it	Gap 3; star-layout design	Boomikaa, Tushar, Saba
5	One record per product and customer: one agreed description per product code, and one customer table with one row per customer number	Gap 2: one clean record per customer or product	Saba, K. Shreya, Vanshika
6	Run quality checks (row counts, empty values, duplicates) and write the results down	Data rules and quality checks	Saba, Chahatleen, Boomikaa
7	Build the dashboard that answers the business decision	Understanding what the business needs	Vanshika, K. Shreya, Chahatleen
8	Write the guide so others can rerun our work, and present it	Understanding what the business needs; Data warehouse basics	Vanshika, Boomikaa, Tushar

4. Gaps closed, success, governance and risks

Gaps closed: (1) data-design tool such as ERWin, learned with a free tool; (2) one clean record per customer or product; (3) designing how data is organised, from drawing to real tables.

Success measures: the pipeline reruns end to end without errors, every quality check passes or is explained, the dashboard answers the business decision, and a classmate can rerun it from the README in one sitting.

Governance: credit the dataset under CC BY 4.0, keep customer number as an anonymous key and never try to identify people, and write down data quality limits in the report.

Risks and fixes: the file is large, so we build on a small sample first. A Databricks Free Edition quota shutdown stops compute for the day, so we avoid looping jobs. We will not use paid tools, live streaming data or scraped websites.

Final recommendation (after the gold tables exist): “Because the data shows [finding], we recommend [action], with the limit that [limit].”